- Equilibrium distribution: The distribution of e^-'s in the conduction band is the product of the density of quantum states, $g_c(E)$ and Fermi-dirac probability function, $f_F(E)$
- Intrinsic semiconductor: no impurities. At T=0K acts as pure insulator. as T↑, we get e^--hole pairs i.e $n=p$
- Intrinsic fermi level: for intrinsic semiconductor, fermi level is very near to middle of bandgap
- Thermal-Equilibrium e^- concentration (n_0)

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{KT}\right]$$

N_c → effective density of states function in conduction band

- Thermal-Eq hole conc (p_0)

$$p_0 = N_v \exp\left[\frac{-(E_F - E_v)}{KT}\right]$$

N_v → effective density of states function in valance band

- Intrinsic carrier concentration (n_i)

$$n_i^2 = N_c \cdot N_v \exp\left[\frac{-E_g}{KT}\right]$$

- Intrinsic fermi-level position $E_{Fi} - E_{midgap} = \frac{3}{4} KT \ln\left(\frac{m_p^*}{m_n^*}\right)$

E_c → conduction band energy (bottom of CB)

E_f → fermi energy (thermal-eq concentrations of e^- & holes)

E_g → bandgap energy : $E_c - E_v$

E_{fi} → intrinsic fermi energy

E_{midgap} → $\frac{E_c + E_v}{2}$

- Donors (n-type): Group V elements (4-covalent bonds)

- Acceptors (p type): Group III elements
- Amphoteric impurities
- Extrinsic semiconductors: $E_f > E_{fi}$ (n-type)
  $E_f < E_{fi}$ (p-type)
- mass action law: $n_0 \cdot p_0 = n_i^2$

$$n_0 = n_i \exp\left[\frac{E_f - E_{Fi}}{KT}\right] \qquad p_0 = n_i \exp\left[\frac{-(E_f - E_{fi})}{KT}\right]$$

- Degenerate semiconductors: if very high doping, impurity states interact & may overlap with the CB/VB. The fermi-level moves into CB (n-type) or VB (p-type) & boltzmann approximations aren't valid
- at typical temperatures, all donor / acceptor atoms are ionized ($T \simeq 300K$)
- at $T = 0K$, no free carriers exist
- $e^{\ominus}$ density in donor state: $\dfrac{N_d}{1 + \frac{1}{2}\exp\left(\frac{E_d - E_f}{KT}\right)}$
- Compensated semiconductor: material having both donor (N_d) and acceptor (N_a) impurities in same region
- $n_0 + N_a^- = p_0 + N_d^+$ (adding donors increases n_0 which decreases p_0 (minority carriers) due to redistribution (annihilation)
- $e^{\ominus}$ conc (complete ionization) (n-type) hole conc (p-type)

$$n_0 = \frac{(N_d - N_a)}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2} \qquad p_0 = \frac{(N_a - N_d)}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$$

- If $N_d \uparrow$ E_f [illegible]

- As $N_d \uparrow$, E_f moves closer to E_c. As $N_a \uparrow$, E_f moves closer to E_v. As $T \uparrow \Rightarrow n_i \uparrow$, E_f moves closer to E_{fi}.
- E_f position in n-type: $E_c - E_f = KT \ln(N_c/N_d)$ $(n_o \approx N_d)$
- " " " p-type: $E_f - E_v = KT \ln(N_v/N_a)$ $(p_o \approx N_a)$
- " relative to intrinsic fermi level: $E_f - E_{fi} = KT \ln(\frac{n_o}{n_i})$
- $N_a \rightarrow$ acceptor conc. $N_d \rightarrow$ donor conc.
  $N_v \rightarrow$ effective density of states function in V.B $\approx 10^{19}$ cm^{-3}
  $N_c \rightarrow$ " " " " " ~ CB $\approx$ " "